

ATLANTIS

A Disaster Simulation Exercise for Training Purposes

by Lieutenant Colonel G.N. Ritchie (Ret'd)

Introduction

This paper examines the process of identifying training needs and methodology and the selection and development of a computer-based simulation exercise as a means of meeting these requirements.

In doing so it seeks to identify guidelines that may be helpful to those involved in similar projects. These guidelines relate to conception, the identification and establishment of modelling parameters, project management and development trials and the integration of the simulation exercise into a study programme.

The paper is based on the experience of the author in a research project which examined the role and responsibilities of the Administration in maintaining life support systems in disaster situations in the developing countries.

The military have used simulations as a means of training for a very long time. At best simulations provide an excellent methodology for training and can highlight the lessons which are the trainer's objectives. At worst they provide an interesting interlude in a course of study but may require careful monitoring to avoid students' developing misconceptions.

Simulations designed for analytical purposes present the same difficulties and dangers although the parameters will be somewhat different.

For purposes of developing the simulation, the following definition of a disaster was used:

A situation resulting from a natural or man-made catastrophe, other than war, demanding total integration of the rescue and life support systems available to the officials responsible for the stricken area, together with the communications and transportation resources required to support the operation.

Identified Areas of Failure in Disaster

Disaster is a highly complex situation in which many influences operate in concert and many in conflict. Some of these factors provided data that are quantifiable but many of them do not.

Clearly, however the role of the local Administration is of paramount importance and the patterns of failures lie in the maintenance of the resources and life-support systems (water, medical aid, food and shelter) and on the transportation and communication systems upon which those life support systems depend.

This helped establish the limits of the models upon which the simulation was to be based. No simulation can hope to represent all the factors and influences present in reality but it is important to isolate those that are most significant and to identify the data that are critical.

It is important to be quite clear about what is to be simulated and the purpose for which the simulation is intended.

The Computer and the Players

Unless the training project to which a computer-based simulation or game relates also aims to illustrate the application of some aspect of computer technology there are good reasons for isolating the players from the computer. We are at some pains to ensure that players do not get the impression that they manage disasters by computer.

Lieutenant Colonel Ritchie works for the Operational Research Branch of the Royal Military College of Science, Shrivenham, Swindon, Wilts, England.

The aim has been to develop an exercise that can be run by a small staff who take it to the players in a variety of institutes and staff colleges in the developing countries. It needs two or three staff assisted by a computer operator and one or two secretary messengers. The computer generates messages in conventional language in response to players' decisions for the deployment of a variety of resources and the level of these resources in the models. It is however equally possible to run this exercise without the assistance of the computer. To do so, however, increases the staff requirement to about 10 or 11.

The Users

The disaster simulation exercise is intended as an element of study programmes for Third World administrators.

In study programmes of this sort the simulation exercise, in which roles are allocated and the areas of responsibility of each role are of significance to the group as a whole, has great potential. It provides an active learning situation and a stimulus to study and discussion.

The Aims and Objectives of the Simulation Exercise

Our aim was to simulate the confused and distorted information flows in emergency situations when there is either a shortage of information or there is far too much of it, but when there is no very effective means of discriminating between the reliable and the false. There is pressure however, upon those with responsibility for making decisions to deploy immediately available resources and to acquire additional ones. Therefore, ATLANTIS is to provide a realistic situation in which the problems and techniques of resource management in disaster can be practised and from which lessons concerning these and the nature and value of disaster preparedness can be drawn.

As already stated, disaster is an extremely complex situation but the original hypothesis concerning the importance of the principal life support systems - rescue, medical aid, water, food and shelter - defined the parameters of our models and of the simulation. As often happens in simulations, the modelling tended to sprout beyond these limits, at one time into fuel and energy generation, clothing and food production and into similar fields. This has been severely limited, to the improvement of the simulation as a whole. The temptation to model too much should be resisted.

Because of the importance of political, economic and sociological factors in disaster we include a player with responsibility for social affairs. His task is to inject the unquantifiable but significant into discussions. This is an important feature; otherwise we run the danger of ignoring many facts of life.

It is important to see ATLANTIS as part of a total teaching package, not as an entity in itself.

The Simulation

The simulation is based upon an imaginary country 'Atlantis', large enough to provide problems associated with more than one level of administrative authority, logistics and transportation, but small enough to be within what I call the 'concept of capability' of the players. In other words, they feel that the problem is of a manageable size.

We chose an imaginary country after some consideration. A real country would have been easier. Players would have the advantage of a ready made understanding, year books, maps, etc. and perhaps a real disaster could have been the basis of the simulation. We rejected this for two reasons. ATLANTIS was designed to train senior members of the administration of developing countries in the planning and management processes related to disaster. By choosing an imaginary setting for the simulation we considered there was more likelihood that our players would think and act objectively. Furthermore, a real situation might prove embarrassing to some users.

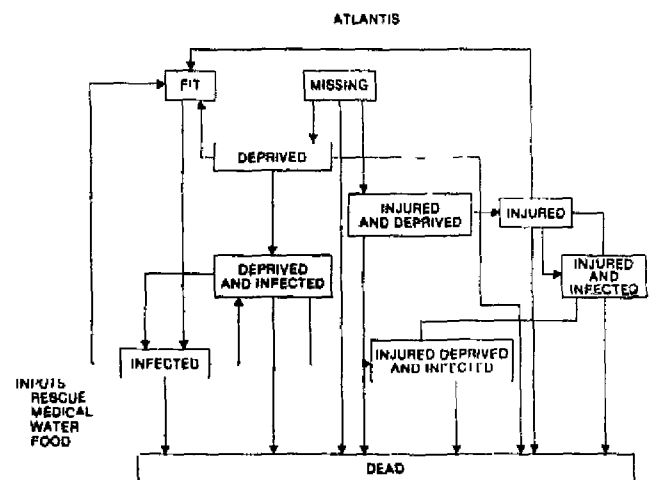


Figure 1. A model of ATLANTIS

Choosing an imaginary country presented certain problems in setting the scene and painting the picture. One must avoid trying to replace what comes from a lifetime of experience with a great mass of pre-exercise reading. There is some evidence to suggest that a sketchy description results in a more commonly held concept, flexible enough to adapt to additional information. What is certain is that the less people have to read, the more likely they are to read it.

ATLANTIS has been designed for up to 20 players, playing singly or in groups as members of the National Disaster Emergency Committee. The exercise is not a game in which various teams are competing against one another. Each group is playing as a team against the situations developed in the simulation.

Comparisons are possible between different teams' performance. In general terms they are aiming to reduce death, sickness and deprivation by the effective and efficient deployment of the various resources available. The fewer dead and sick and the more water and food available to the survivors, the more effective one can say has been the Disaster Committee's work.

Figure 1 illustrates one of the ATLANTIS models. Figure 2 demonstrates two ways of assessing player performance.

Setting up the Exercise

Before taking part in the exercise, players are required to read material which introduces them to ATLANTIS, its topography, economy and people, plus a detailed description of their own roles, responsibilities and resources and the playing instructions. Only the Prime Minister and the disaster coordinator, as chairman of the committee, have details of each player's role, responsibilities and resources. All have a general idea of what the other players are responsible for but they do not have details. We have thus restricted pre-exercise reading to about one-and-a-half hours. The Prime Minister and disaster co-ordinator need more time.

The exercise is not a training course in its own right but an important element of a total teaching package. The opportunity therefore exists for players to be briefed before the exercise when preliminary instructions are issued. In some cases, however, players have met the exercise director and one another only when they have sat down to take part.

The more time available for pre-exercise briefing, the more effective the players' action is likely to be. There are, of course, training advantages to be drawn from illustrating how communication breakdowns exist and occur in groups even when all have a common aim. It can also be shown how important the role of the chairman is in ensuring that information is shared and that the deployment of resources is coordinated.

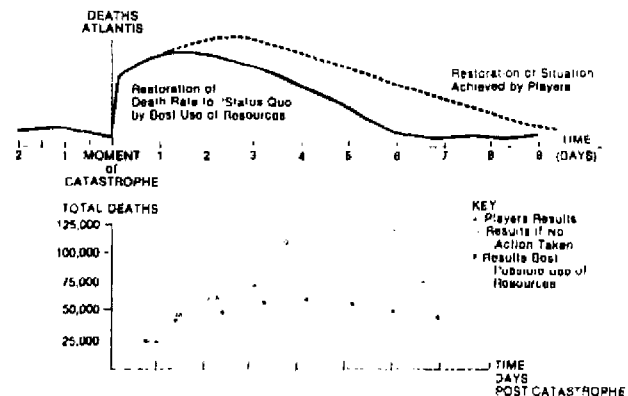


Figure 2. Two ways of showing player performance

Structure of the Simulation Exercise

When ATLANTIS is used as an element of a training course it may be used on more than one occasion to practise and illustrate various aspects of the programme of studies, and to this end the simulation has been designed in three separate, but overlapping, phases. The exercise may be run either straight through from start to finish and when used in this way requires from three to four working days. Alternatively, the exercise may be run to illustrate the problems and aspects of a specific phase over shorter periods. The details of the phases are:

Phase 1. This is the period immediately after the catastrophe during which players are collecting and collating the information that will enable them to assess the situation and its requirements. They are also establishing the location and availability of various resources required to meet the needs of the disaster situation.

Phase 2. Sees the continuation of Phase 1 and the deployment of the various forms of rescue, relief and repair resources.

Phase 3. Introduces the problems associated with the logistics and the integration of incoming foreign relief with national resources.

No one phase is unique however, and each contains elements of the problems that arise throughout the exercise.

The unquantifiable social problems are introduced throughout. In addition we introduce a variety of extraneous subjects and interference to the committee. These take the form of enquiries and requests concerning a variety of subjects, for example the fate of foreign nationals, the necessity to brief a high ranking UN official, and many more.

Real and Simulated Time

There is the problem of simulated time. During Phase 1 the players operate in real time and the directing staff, with computer backing, operates in simulated time. We see no advantage in frustrating the players by leaving them to wait around for replies to their requests for information or reports from the reconnaissance effort that they have deployed. The directing staff, aided by computer, responds immediately and tries to give more information than players can handle rather than starving them of it.

As we move into Phase 2 however, we introduce simulated time with distinct playing periods representing 24 hours of real time. As Phase 2 develops the playing period is reduced. This has been done for two reasons: pressure on players is high during Phase 1 but begins to lighten during Phase 2, particularly around days five and six. Unless an unrealistically high rate of disease infection, with an equally artificially short period of incubation were introduced, players would not experience the consequences of certain failures in preventive action during early days of play.

Observers

To compensate for the pace of the exercise and to ensure that important matters that arise during the play but are not fully or adequately discussed are, in fact, discussed, it is important to include an observer throughout the play.

The observer, a member of the directing staff, is not a critical monitor of players' actions but a link between the exercise and the programme of studies of which it is a part. Additionally he links the players with the directing staff, and can alert the directing staff to any problem or misunderstanding that is likely to affect play.

Conclusions

The development of ATLANTIS drew the following lessons:

- The development of any simulation or game must be based upon a thorough programme of research or have an adequate depth of knowledge and experience immediately available to it.
- Those involved, both user and developer, must be quite clear from the outset about:
 - the nature and parameters of the models and the stimulation;
 - the purpose of the simulation, exercise or game; and
 - for whom the simulation, exercise or game is intended.
- Development must include a programme of consultation and presentations between user and developer and a series of development trials.

Acknowledgement

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